



Spatial heterogeneity of microscopically detected *Nosema* spp. infection in honeybees: Climate-associated patterns in the Eastern Black Sea Region of Türkiye

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ARTICLE INFO

Keywords:

Apis mellifera
Climate change
Humidity
Spatial heterogeneity
Temperature

ABSTRACT

Nosemosis has emerged as the predominant microsporidian infection of honeybees (*Apis mellifera* L.) in many regions, including Türkiye, yet the climatic drivers of its epidemiology remain insufficiently quantified. This study assessed the effects of temperature and relative humidity on nosemosis prevalence across seven provinces in the climatically diverse Eastern Black Sea region. A dataset of 2688 colony-level observations was analyzed using generalized linear models, mixed-effects models, beta regression, generalized additive models, and spatial autocorrelation. Both temperature and humidity were consistently significant predictors ($p < .001$). Quadratic temperature terms and nonlinear GAM smooths were consistent with a thermal performance pattern, with infection peaking at intermediate ranges. Humidity exerted threshold-like effects, with disproportionately higher prevalence at upper quantiles. Province-level random effects accounted for ~18% of variance, underscoring spatial heterogeneity, while elevation was not significant once climatic variables were included. Exploratory spatial analysis (Global Moran's $I = -0.476$, $p = 0.043$) tentatively suggested dispersion rather than clustering; however, given the very small number of spatial units ($n = 7$), this result should be regarded as hypothesis-generating only. The beta mixed-effects model achieved moderate predictive accuracy under cross-validation (RMSE ≈ 0.109). Collectively, these findings indicate that nosemosis prevalence in the region is strongly associated with climatic variability, exhibits nonlinear and threshold-like responses, and shows pronounced spatial heterogeneity. We emphasize that these are statistical associations and that unmeasured factors (e.g., colony strength, nutrition, pesticide exposure, management practices, and co-infections) likely contribute to the unexplained variance. By integrating hierarchical, nonlinear, and spatial approaches, this study establishes a climate-associated epidemiological assessment of nosemosis risk in *A. mellifera* and highlights the importance of localized climatic suitability under ongoing climate change.

1. Introduction

The western honeybee (*Apis mellifera*) is a globally vital pollinator that supports biodiversity and crop production. However, honeybee health has been declining worldwide because of the combined effects of pathogens, environmental stressors, and climate variability (Goulson et al., 2015; Potts et al., 2010). Among the pathogens threatening honeybees, microsporidian parasites of the genus *Nosema* cause a disease known as nosemosis, which infects the gastrointestinal tract of adult bees and is associated with reduced longevity, impaired colony performance, energetic stress, and compromised immunity (Li et al.,

2018; Ostap-Chec et al., 2024). There are two principal etiological agents of nosemosis: *Nosema apis* and *Nosema ceranae* (Özgör et al., 2015).

Since its detection in *Apis mellifera*, *Nosema ceranae* has become globally dominant in many regions (Martín-Hernández et al., 2018). The global distribution and increasing prevalence of *N. ceranae* are of particular concern because this pathogen exhibits greater thermotolerance and environmental persistence compared to *N. apis* (Fenoy et al., 2009). Experimental evidence also suggests that *N. ceranae* infections can alter honeybee physiology and behavior, including early maturation of nurse bees and disruption of colony balance, with potential links to

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<https://doi.org/10.1016/j.vetpar.2026.110835>

Received 13 April 2026; Received in revised form 19 June 2026; Accepted 20 June 2026

Available online 25 June 2026

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colony collapse phenomena (Li et al., 2018).

The epidemiology of nosemosis is influenced by biotic and abiotic factors, including season, climate, and landscape context. Seasonal increase in temperature and humidity have been correlated with higher infection prevalence, suggesting that climatic conditions may influence both pathogen viability and transmission dynamics (Jabal-Uriel et al., 2022; Ostap-Chec et al., 2024). Cross-regional surveys indicate considerable spatial heterogeneity in nosemosis prevalence, with geographic and climatic gradients contributing to variance in infection levels (Ansari et al., 2017; Özgör et al., 2015). These findings implicate climate variables as key ecological determinants of disease risk. However, the precise nature of these relationships remains underexplored in many regions.

In Türkiye, early molecular detection confirmed the presence of *N. ceranae* in honeybee colonies, and prevalence studies have documented its widespread occurrence, often with positive correlations between humidity and spore abundance (Ütük et al. 2016; Ütük et al., 2010). For example, research in Kırşehir found significant positive associations between relative humidity and *Nosema* spore counts, and molecular analyses indicated that *N. ceranae* was the sole pathogen detected (Özgör et al., 2015; Ütük et al., 2010). Moreover, climatic and geographic differences have been shown to affect *Nosema* incidence, suggesting that environmental modulators play an important role in shaping infection risk across different Turkish regions (Özgör et al., 2015).

Despite these regional insights, there remains a lack of integrated epidemiological models that jointly quantify climatic drivers, hierarchical structure, nonlinear responses, and spatial autocorrelation in nosemosis dynamics at the scale of climatic zones, such as the Eastern Black Sea region. This is important because honeybee colonies in this region experience pronounced climatic heterogeneity due to altitudinal gradients and maritime influences, which could profoundly influence disease ecology.

To address this gap, we examined the climatic correlates of nosemosis prevalence across seven provinces in the Eastern Black Sea region using a suite of statistical approaches, including generalized linear models, mixed-effects models, beta regression, generalized additive models, and spatial autocorrelation analysis. By integrating nonlinear response curves and hierarchical spatial structure, we aimed to characterize the climatic correlates of infection prevalence and to develop a quantitative epidemiological assessment of climate–nosemosis associations. This work contributes to a growing understanding of how environmental variation influences pathogen dynamics in managed honeybees and offers insights relevant to apicultural management in the context of ongoing climate change. Throughout the manuscript, we use the term “infection prevalence” in its epidemiological sense, namely the proportion of microscopically infected adult bees per colony-level sample at a single time point; this single, clearly defined response variable is used in place of related but non-equivalent terms such as “incidence” (which would require explicit time-at-risk denominators), “infection risk” (a predicted probability), or “spore load/abundance” (a continuous intensity measure that would require quantitative spore counts).

On the basis of prior field and laboratory evidence reviewed above, we evaluated four explicit hypotheses regarding the climatic and spatial epidemiology of microsporidian infection in the study system: (H1) relative humidity is positively associated with infection prevalence, with a non-linear, threshold-like response at the upper end of the humidity gradient; (H2) temperature is non-linearly associated with infection prevalence, with prevalence peaking at intermediate values consistent with a thermal performance pattern, rather than monotonically increasing or decreasing across the gradient; (H3) after climatic effects are accounted for, residual variation in infection prevalence is hierarchically structured among provinces and across months, rather than being homogeneously distributed; and (H4) province-level infection prevalences are not strongly positively spatially autocorrelated at the

provincial scale, but instead reflect local climatic suitability, such that the spatial autocorrelation analysis is regarded as exploratory and hypothesis-generating rather than confirmatory. Hypothesis H4 was not formulated post hoc but follows directly from earlier multi-year and cross-regional surveys, which have repeatedly documented that nosemosis prevalence varies idiosyncratically among neighbouring areas and tracks local climatic and management conditions rather than forming spatially contiguous high-prevalence clusters (Gisder et al., 2010; Özgör et al., 2015; Ütük et al. 2016). On this prior basis, the absence of strong positive spatial autocorrelation at the provincial scale was an a priori expectation rather than a description of the observed result. Each hypothesis is evaluated using the corresponding component of the integrated modelling pipeline described in the Materials and Methods.

2. Materials and methods

2.1. Study area and sampling design

The study was conducted in 14 localities in seven provinces of the Eastern Black Sea region of Türkiye (Artvin, Rize, Trabzon, Giresun, Ordu, Samsun, and Sinop), a geographically heterogeneous area characterized by strong altitudinal gradients, high annual humidity, and maritime climatic influence. Colonies were selected following defined inclusion and exclusion criteria. Inclusion criteria were: (i) overwintered, queen-right colonies of *A. mellifera* maintained in Langstroth-type hives by local beekeepers, (ii) colonies present at the same apiary for the entire four-year sampling period to ensure repeated-measures continuity, and (iii) absence of acute mortality, queen loss, or hive abandonment at the time of sampling. Exclusion criteria were: (i) colonies treated with anti-microsporidian compounds (e.g., fumagillin) within the four weeks preceding any sampling event, (ii) recently introduced packages or swarms (< 3 months) for which baseline infection status was undefined, and (iii) colonies that collapsed or were merged during the sampling period, in which case the corresponding monthly cell was treated as missing. Sampling was conducted monthly over a four-year period (January 2022–December 2025), with a total of 2688 colony-level observations (14 localities × 12 months × 4 replicates × 4 years) (see [Supplementary File S1](#) for the complete observational dataset). For each sampling event, four biological replicates – here defined as four independent colonies sampled per locality per month, with each colony providing a single composite pool of ~30 adult forager bees collected from the hive entrance – were obtained from managed *A. mellifera* colonies maintained by local beekeepers to quantify nosemosis prevalence. Infection prevalence was defined as the proportion of microscopically infected adult bees per sample of approximately 30 bees, following the standard protocol (Fries et al., 2013). Because the diagnosis was based on light microscopy of spore preparations, infection refers to detection of *Nosema* spp. spores in the host midgut; the microscopic method employed here does not discriminate between *N. ceranae* and *N. apis*, nor does it formally identify coinfections. Accordingly, the response variable should be interpreted as prevalence of microsporidian (*Nosema* spp.) infection rather than as species-specific *N. ceranae* prevalence. Infection prevalence was expressed as the percentage of infected individuals per sample, and this terminology (“infection prevalence”) is used consistently throughout the manuscript in place of the previously used “infection rate.”

Geographic coordinates (latitude and longitude) and elevation values were recorded for each locality to support spatial and hierarchical analyses. Thus, the dataset had a nested structure consisting of repeated monthly observations within localities nested within provinces. The decimal-degree coordinates of each of the 14 sampling localities (apiaries), together with elevation in metres above sea level, are provided in full in [Supplementary File S1](#) to enable independent replication and to allow future researchers to relocate the exact apiary sites. The precise geographic positions of all sampled apiaries are additionally mapped in [Supplementary Fig. 1](#), allowing future researchers to track

the exact location of each apiary beyond the generalized province-level boundaries shown in the main-text maps.

2.2. Climatic variables

Mean monthly temperature (°C) and relative humidity (%) were used as climatic predictors. Monthly mean values were obtained from the Turkish State Meteorological Service (TSMS) weather station records corresponding to the nearest station within each province. Stations were selected based on spatial proximity and data completeness; station-to-locality distances ranged from 4.2 to 28.6 km. For localities lacking a proximal station, temperature and humidity values were spatially interpolated using inverse distance weighting from the two nearest stations. These variables were selected based on prior evidence demonstrating their influence on *Nosema* proliferation, spore viability, and transmission dynamics (Fenoy et al., 2009; Gisder et al., 2010). Climatic variables were treated as continuous predictors in baseline models and further evaluated for nonlinear and threshold effects. The Pearson correlation between temperature and relative humidity across the dataset was $r = -0.41$ ($p < .001$), indicating a moderate negative association. Variance inflation factors (VIF) for temperature (VIF = 1.20) and humidity (VIF = 1.20) were both well below the commonly applied threshold of five, confirming the absence of problematic multicollinearity (Zuur et al., 2009).

2.3. Data preparation

The outcome variable (infection prevalence, as defined in the Study Area and Sampling Design) represents sample-level infection prevalence rather than population-level disease prevalence *sensu stricto*; the term “infection prevalence” is used throughout to denote this sample proportion, consistent with usage in the *Nosema* monitoring literature (Fries et al., 2013; Gisder et al., 2010). Because infection rates represent proportional data bounded between 0% and 100%, values were rescaled to the (0,1) interval prior to beta regression modeling. A small constant (0.001) was added before scaling to avoid boundary issues, following recommendations for beta regression implementation (Cribari-Neto and Zeileis, 2010). The value 0.001 was chosen because it is small enough relative to the observed range of infection prevalences that it does not appreciably shift the central tendency of the distribution, while still being large enough to displace exact 0 and 1 values away from the closed beta boundary at machine-precision tolerance. Sensitivity analyses in which the constant was varied between 0.0005 and 0.005 yielded fixed-effect estimates that differed by $< 1\%$ on the logit scale, confirming that the choice of constant did not materially influence the reported inference.

Data exploration procedures followed the best-practice guidelines for ecological modeling (Zuur et al., 2009), including checks for collinearity and distributional properties.

2.4. Sequential modelling pipeline and inferential status

Statistical models were applied in a deliberately sequential pipeline in which each successive step relaxes one assumption violated by the preceding step, so that the unique contribution of each model is interpretable. (i) The Gaussian GLM serves as an exploratory baseline establishing the direction and approximate magnitude of climatic associations. (ii) Linear mixed-effects models (LMM, with quadratic temperature) introduce hierarchical (province- and month-level) structure to partition variance and to test for symmetric, low-order non-linearity in the temperature response (H2, H3). (iii) Beta regression and beta mixed-effects regression replace the Gaussian assumption with a distributionally appropriate model for proportional outcomes bounded on (0,1), providing the primary inferential framework for fixed-effect estimates and their uncertainties. (iv) Generalized additive models (GAM) and generalized additive mixed models (GAMM) allow flexible, data-

driven non-linear smooths of climatic predictors and are used to test whether the LMM's quadratic specification adequately captures the realized response shapes, and to evaluate how much of the apparent non-linearity is absorbed by hierarchical random effects (H2 vs H3 interaction). (v) Predictive accuracy of the beta mixed model is then assessed by k-fold cross-validation. (vi) Finally, Global Moran's I is used to evaluate H4 in an explicitly exploratory mode, given the limited number of spatial units. The GLM, LMM, GAM/GAMM and Moran's I steps are exploratory or comparative; the beta mixed-effects model is the principal confirmatory inferential model. To explicitly account for temporal structure beyond the inclusion of “Month” as a random intercept, we (a) inspected autocorrelation functions (ACF) of the model residuals at multiple lags, (b) fitted an additional GAMM specification with a cyclic cubic-regression smooth $s(\text{Month}, \text{bs} = "cc")$ to capture seasonal cyclicity, and (c) examined whether the inclusion of one- and three-month lagged climatic predictors improved fit. None of these diagnostics indicated material residual serial autocorrelation beyond that already absorbed by the hierarchical random effects, and the seasonal cyclic smooth did not change the sign or significance of the main climatic fixed effects; full diagnostic output is provided in [Supplementary File S2](#).

2.4.1. Baseline linear modeling (GLM)

A Gaussian generalized linear model (GLM) was first fitted as an exploratory baseline to evaluate the linear relationships between infection prevalence and climatic predictors. Because infection prevalence is a proportion bounded between 0 and 1, the Gaussian GLM is not the distributionally appropriate framework for inference; it is reported here solely as a first-stage exploratory benchmark to establish the direction and approximate magnitude of climatic associations prior to the implementation of distributionally appropriate hierarchical (mixed-effects) and beta regression models, which constitute the primary inferential framework. Generalized linear models extend classical linear models and allow flexible error structures (Zuur et al., 2009). Model fit was assessed using Akaike's Information Criterion (AIC) following an information-theoretic framework (Burnham and Anderson, 2002).

Model structure:

$$\text{Infection} \sim \text{Temperature} + \text{Humidity}$$

2.4.2. Linear mixed-effects modeling (LMM)

To account for hierarchical spatial and temporal structures, linear mixed-effects models were fitted using the lme4 package (Bates et al., 2015). Province and month were included as random intercepts:

$$\text{Infection} \sim \text{Temperature} + \text{Humidity} + (1 \mid \text{Province}) + (1 \mid \text{Month})$$

Variance components were extracted to calculate the intraclass correlation coefficient (ICC), which quantifies the proportion of variance attributable to province-level effects (Nakagawa and Schielzeth, 2013).

2.4.3. Elevation analysis

Elevation was incorporated as an additional fixed effect to test whether altitude independently explained infection variability beyond climatic drivers. Its statistical significance was evaluated using a mixed-effects framework.

2.4.4. Quadratic temperature modeling

To evaluate the potential thermal performance patterns, a quadratic temperature term was added:

$$\text{Infection} \sim \text{Temperature} + \text{Temperature}^2 + \text{Humidity} + (1 \mid \text{Province}) + (1 \mid \text{Month})$$

Quadratic modeling is commonly used to detect unimodal or optimal responses in ecological systems (Crawley, 2012).

2.4.5. Humidity threshold analysis

To test for nonlinear threshold effects, relative humidity was categorized into quantile-based groups. These categorical predictors were included in the mixed-effects models to evaluate stepwise increases in infection risk.

2.4.6. Beta mixed-effects regression

As infection rates are proportional outcomes, beta mixed-effects models were fitted using the glmmTMB package (Brooks et al., 2017). The beta distribution is appropriate for modeling continuous proportions constrained between 0 and 1 (Cribari-Neto and Zeileis, 2010).

Model structure:

$\text{Infection}_{\text{beta}} \sim \text{Temperature} + \text{Humidity} + (1 \mid \text{Province}) + (1 \mid \text{Month})$

Random-effect variance components were examined to assess hierarchical spatial influence.

2.4.7. Generalized additive models (GAM)

To investigate nonlinear climatic responses, generalized additive models (GAMs) were fitted using penalized regression splines (Wood, 2017). Smooth functions were applied to temperature and humidity as follows:

$\text{Infection} \sim s(\text{Temperature}) + s(\text{Humidity})$

The effective degrees of freedom (edf) were used to quantify nonlinearity.

2.4.8. Generalized additive mixed models (GAMM)

Generalized additive mixed models were fitted to integrate nonlinear response modeling with a hierarchical structure. Random intercepts for province and month were included alongside smooth climatic terms, allowing the separation of nonlinear climatic effects from spatial heterogeneity (Wood, 2017).

2.4.9. Model comparison

Model comparison was conducted within an information-theoretic framework using Akaike's Information Criterion (AIC) (Burnham and Anderson, 2002). Because candidate models were fitted under different likelihood families (Gaussian vs. beta) and, in some cases, different effective sample sizes, AIC values were not interpreted as strictly comparable across all model classes. Instead, AIC comparisons were restricted to models sharing identical likelihood structures. Differences between Gaussian mixed models and beta mixed-effects models were therefore evaluated in terms of distributional appropriateness, variance partitioning, explanatory performance, and predictive accuracy rather than absolute AIC magnitude. This approach avoids invalid cross-family comparisons and ensures methodological rigor. A summary of the model fit statistics is presented in Table 1.

2.4.10. Cross-validation

The predictive performance of the beta mixed-effects model was evaluated using five-fold cross-validation. Root Mean Square Error (RMSE) was calculated to quantify out-of-sample prediction error. Cross-validation procedures followed standard predictive modeling

practices (James et al., 2021). No systematic over- or under-prediction bias was observed across the validation folds, indicating stable model performance.

2.4.11. Spatial analysis

The spatial structure of the model-predicted infection risk was assessed using Global Moran's I statistic. Spatial weights were constructed using province-level adjacency relationships. Moran's I quantifies spatial autocorrelation and was interpreted using a randomization framework (Bivand and Wong, 2018; F. Dormann et al., 2007).

Local indicators of spatial association (LISA) were not performed because of the limited number of spatial units ($n = 7$), which constrains reliable local-level spatial inference.

Negative Moran's I values indicate spatial dispersion, whereas positive values indicate clustering. Because the analysis was restricted to $n = 7$ provinces, the statistical power for spatial inference is intrinsically low; the Moran's I analysis was therefore explicitly conducted as an exploratory, hypothesis-generating procedure rather than as a confirmatory spatial test, and no ecological conclusions regarding spatial structure are drawn solely from this statistic.

Software

All analyses were conducted using R version 4.4.1. The following packages were used:

- lme4 for mixed-effects modeling (Bates et al., 2015)
- glmmTMB for beta regression (Brooks et al., 2017)
- mgcv for GAM and GAMM modeling (Wood, 2017)
- spdep for spatial autocorrelation analysis (Bivand and Wong, 2018)
- caret for cross-validation procedures (Kuhn, 2008)

3. Results

Nosemosis rates exhibited marked spatial variation across the study region. Provinces characterized by higher relative humidity and moderate temperature regimes showed elevated infection levels, whereas comparatively drier regions displayed lower prevalence (Fig. 1). This spatial heterogeneity suggests a potential climatic gradient influencing disease dynamics.

The baseline generalized linear model (GLM) revealed strong positive effects of both relative humidity ($\beta = 0.56$, $p < .001$) and temperature ($\beta = 1.09$, $p < .001$) on infection rates. The inclusion of climatic predictors substantially reduced the model deviance relative to the null model, indicating that temperature and humidity explain a considerable proportion of the variability in infection prevalence. However, this model does not account for hierarchical spatial and temporal structures.

When province and month were incorporated as random intercepts in a linear mixed-effects model, both climatic predictors remained statistically significant (humidity: $t = 2.96$; temperature: $t = 4.21$). Variance partitioning revealed that province-level variability ($\sigma^2 = 24.47$) exceeded month-level variability ($\sigma^2 = 2.47$), with residual variance estimated at 91.57. The intraclass correlation coefficient indicated that approximately 18% of the total variance was attributable to differences among provinces, indicating a moderate level of spatial heterogeneity among provinces and leaving the majority of variance unexplained by

Table 1
Information-theoretic comparison of candidate models.

Model	Distribution	Random Effects	AIC	Notes
GLM	Gaussian	None	8391.27	Baseline linear model
LMM	Gaussian	Province + Month	2088.53	Hierarchical structure included
Beta mixed model	Beta	Province + Month	−958.31	Appropriate for proportional data
GAM	Gaussian	None	7993.04	Nonlinear smooth terms
GAMM	Gaussian	Province + Month	2068.82	Nonlinear + hierarchical

Note: AIC values are directly comparable only among models fitted under the same likelihood family and response scaling. Cross-family comparisons (Gaussian vs. beta) were provided for descriptive interpretation only.

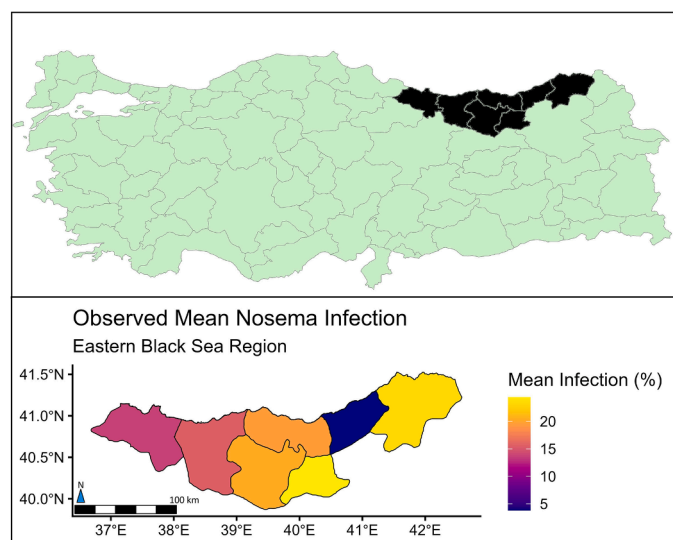


Fig. 1. Mean nosemosis prevalence (%) across seven provinces in the Eastern Black Sea region of Türkiye. Color gradient: blue = lowest prevalence, yellow = highest prevalence (sequential viridis scale). Province labels and apiary sampling sites are overlaid on the map; administrative boundaries are shown in black.

province-level structure alone.

Subsequently, elevation was introduced as a fixed effect in the mixed model; however, it was not statistically significant, suggesting that climatic variables exerted their effects on infection prevalence largely independently of elevation itself. In other words, once temperature and relative humidity were included in the model, the residual contribution of elevation as a stand-alone topographic variable was no longer detectable, consistent with the interpretation that the influence of altitude on infection is mediated by its effect on local climatic conditions rather than by altitude as a direct physical predictor.

To evaluate the potential nonlinear temperature responses, a quadratic temperature term was fitted. The quadratic component was statistically significant, providing evidence for a nonlinear

temperature–infection relationship (quadratic-term coefficient and standard error reported in [Supplementary File S2](#)). Infection prevalence increased with temperature to an intermediate range, after which marginal increases attenuated. The ecological interpretation of this pattern in relation to thermal performance theory is reserved for the Discussion.

Nonlinear humidity effects were further explored by categorizing relative humidity into quantile-based groups. Compared with the lowest humidity category, mid-range humidity levels were associated with significantly higher infection rates ($\beta \approx 9.25$, $p < .001$), and the highest humidity group exhibited an even stronger increase ($\beta \approx 15.22$, $p < .001$). These results indicate a threshold-like response to increasing humidity.

Because infection rates represent proportional data, a beta mixed-effects regression model was implemented. Both humidity ($z = 4.62$, $p < .001$) and temperature ($z = 5.13$, $p < .001$) remained highly significant predictors on the logit scale. Province-level random variance (0.100) exceeded month-level variance (0.013), reinforcing the importance of the spatial structure. The predicted infection probabilities increased steadily with climatic intensity ([Fig. 2](#)).

Generalized additive modeling revealed strong nonlinear responses for both humidity (edf = 8.75, $p < .001$) and temperature (edf = 7.44, $p < .001$). The GAM explained 53.3% of the total deviance (adjusted $R^2 = 0.526$), exceeding the explanatory power of the linear models. Response curves ([Fig. 3](#)) demonstrated accelerated infection increases at higher humidity levels and a nonlinear temperature-dependent pattern consistent with ecological thermal performance curves. However, when the hierarchical spatial structure was incorporated through a generalized additive mixed model (GAMM), the effective degrees of freedom were reduced to approximately 1 for both predictors. This reduction suggests that part of the apparent nonlinearity observed in the pooled data was attributable to province-level heterogeneity rather than purely functional climatic responses.

Model comparison indicated that hierarchical and distributionally appropriate models provided an improved fit relative to the baseline GLM. Although direct AIC comparison across different likelihood families must be interpreted cautiously, both the beta mixed-effects model and GAMM demonstrated greater parsimony and explanatory realism than simple linear approaches.

The model-predicted spatial risk surface closely matched the

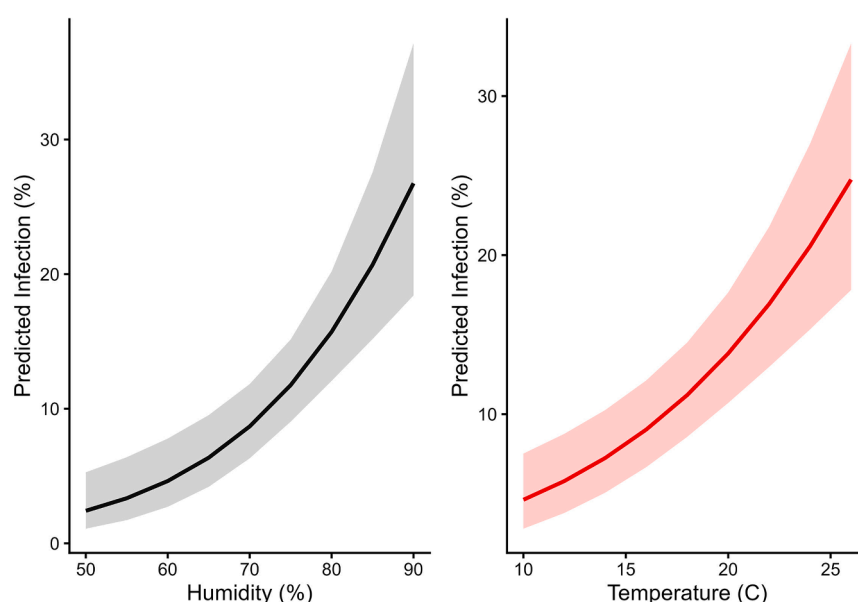


Fig. 2. Predicted nosemosis probability (%) across the observed gradient of (A) relative humidity and (B) temperature, based on the beta mixed-effects regression model. Predictions were generated while holding other predictors constant at their mean values and excluding random effects (population-level predictions). The curves illustrate an increasing infection risk with rising humidity and temperature, consistent with the fixed-effect estimates reported in the main analysis.

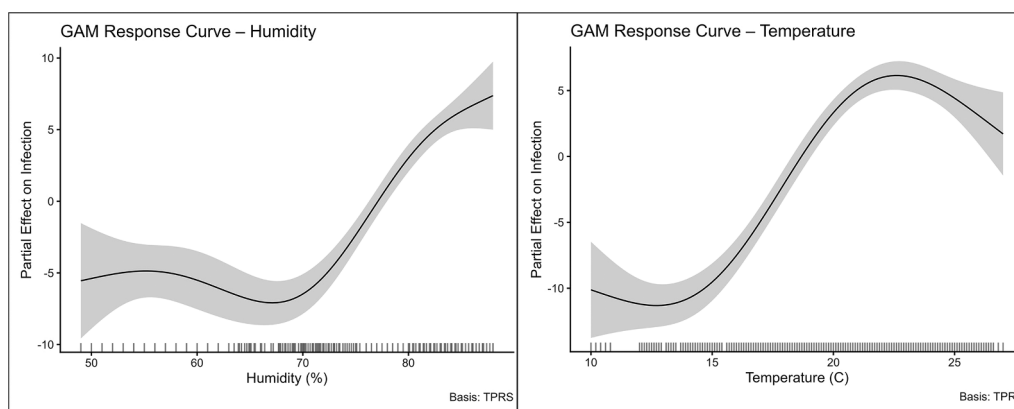


Fig. 3. Smoothed response functions showing the nonlinear effects of (A) relative humidity and (B) temperature on nosemosis prevalence derived from the generalized additive model (GAM, without random effects). The shaded areas represent 95% confidence intervals. The effective degrees of freedom (edf) indicate significant nonlinearity for both predictors in the GAM framework. Caution is warranted in the interpretation of these smooths, as the effective degrees of freedom were reduced to approximately 1 in the generalized additive mixed model (GAMM) after accounting for province-level random effects, suggesting that part of the apparent nonlinearity may reflect spatial heterogeneity among provinces rather than purely functional climatic responses.

observed infection distribution (Fig. 4). Provinces with elevated humidity and moderate temperature regimes exhibited the highest predicted infection risk, whereas drier regions had lower probabilities.

Exploratory spatial analysis using Global Moran's I yielded a negative value ($I = -0.476$, $p = 0.043$; Fig. 5). Because this statistic was computed across only seven provincial units, its statistical power is intrinsically limited and the result must be regarded strictly as hypothesis-generating. Accordingly, we do not draw ecological conclusions regarding spatial dispersion versus clustering on the basis of this analysis; finer-scale spatial data (e.g., locality- or apiary-level) and dedicated spatial regression frameworks will be required to test whether such a pattern is robust.

The five-fold cross-validation of the beta mixed-effects model yielded a mean RMSE of 0.109 (approximately 10.9% on the percentage scale), indicating moderate predictive accuracy under out-of-sample validation.

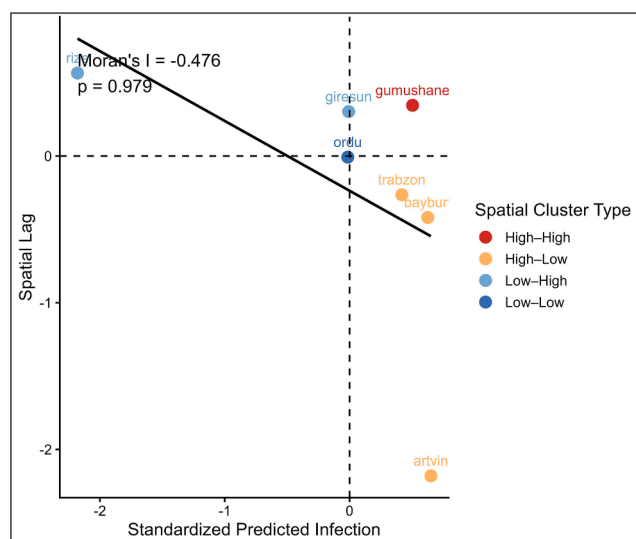


Fig. 5. Global Moran's I correlogram for province-level predicted nosemosis prevalence. The observed Moran's I value ($I = -0.476$, $p = 0.043$) suggests negative spatial autocorrelation, tentatively indicating spatial dispersion rather than clustering of infection risk across provinces. Given the limited number of spatial units ($n = 7$), this result should be interpreted as exploratory rather than confirmatory. Because of the low statistical power inherent to $n = 7$ spatial units, this analysis is reported as hypothesis-generating only; no ecological inference regarding contagion or environmental filtering should be drawn solely from this figure.

4. Discussion

This study provides a comprehensive, climate-integrated assessment of nosemosis dynamics in honeybee (*A. mellifera*) colonies in the Eastern Black Sea region of Türkiye. By combining hierarchical mixed-effects modeling, beta regression, nonlinear GAM/GAMM approaches, spatial autocorrelation analysis, and cross-validation, we show that infection prevalence is strongly associated with climatic conditions, exhibits nonlinear and threshold-like responses, and displays pronounced spatial heterogeneity. We emphasize that these associations should not be interpreted causally, as other established determinants of honeybee health (e.g., colony strength, nutrition, pesticide exposure, beekeeping practices, and co-infections with viruses or *Varroa destructor*) were not measured in the present study and may contribute substantially to the

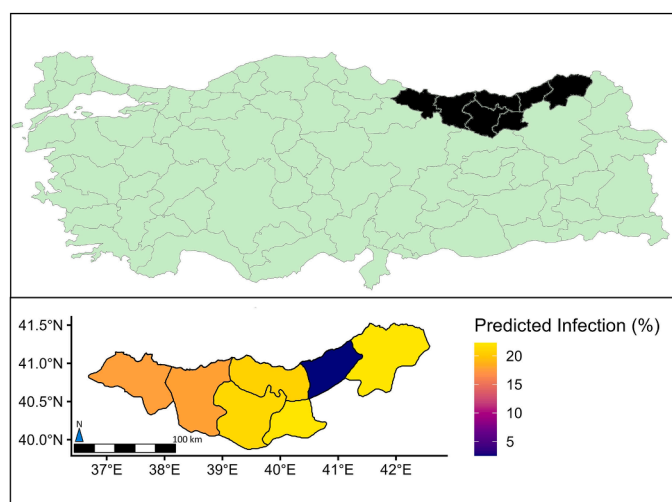


Fig. 4. Province-level predicted nosemosis prevalence (%) derived from the beta mixed-effects regression model, including temperature and relative humidity as fixed effects and province and month as random intercepts. Color gradient: blue = lowest predicted prevalence, yellow = highest predicted prevalence (sequential viridis scale). Province labels and apiary sampling sites are overlaid on the map; administrative boundaries are shown in black.

unexplained variance.

4.1. Methodological considerations

During five-fold cross-validation of the beta mixed-effects model, convergence warnings related to non-positive-definite Hessian matrices were observed in multiple validation folds. Such warnings arise in mixed-effects beta regression when within-level sample sizes are small or when variance components approach boundary estimates (Brooks et al., 2017). To assess whether these warnings reflected genuine instability, the affected folds were re-fitted using alternative optimizers (bobyqa and nlminbwrap via the optimx package), and the parameter estimates were compared across optimizers. The fixed-effect estimates for temperature and humidity showed negligible differences across optimizers, and the RMSE values were consistent across validation folds, indicating no systematic over- or under-prediction. The province-level variance component remained consistently small in the affected folds, suggesting that the boundary condition arose from the limited number of random-effect levels per fold rather than fundamental model misspecification. Nonetheless, these warnings indicate that the current model operates near its sample-size limits for hierarchical beta regression, and future studies with a larger number of provinces and longer sampling periods should explicitly test whether increased spatial and temporal replication eliminates convergence issues.

4.2. Climatic drivers and field consistency

Both temperature and relative humidity emerged as robust predictors across all statistical frameworks. The consistency of these effects in GLM, LMM, beta mixed models, and GAM analyses indicates that the observed associations are not artifacts of model structure but reflect underlying ecological processes.

Laboratory experiments have shown that *Nosema ceranae* spores exhibit high resistance to temperature variation and desiccation, enhancing environmental persistence (Fenoy et al., 2009). Moreover, comparative virulence studies have demonstrated that *N. ceranae* differs from *N. apis* in terms of thermal tolerance and epidemiological behavior (Forsgren and Fries, 2010). These experimental findings are consistent with our field-based observation that temperature is significantly associated with infection prevalence, though we caution that field-scale associations cannot, in themselves, isolate the direct physiological effects of temperature from co-varying climatic or management factors.

Humidity showed particularly strong effects and displayed threshold-like behavior. Infection risk increased disproportionately at higher humidity levels, suggesting that moisture facilitates spore survival and transmission. Similar associations between elevated humidity and higher nosemosis prevalence have been documented in longitudinal European studies (Gisder et al., 2010). In Türkiye, molecular surveillance studies have confirmed the widespread dominance of nosemosis originated from *N. ceranae* across multiple provinces, indicating that climatic suitability likely contributes to its persistence and spread (Ütük et al. 2016).

In Türkiye, multiple regional studies have documented the widespread prevalence of nosemosis, often identifying it as the dominant apian disease across diverse climatic zones (Özgör et al., 2015; Ütük et al. 2016). Seasonal surveys conducted in Central Anatolia and the Mediterranean regions have similarly reported strong associations between humidity, seasonal transitions, and spore prevalence (Emsen et al., 2020). These findings collectively indicate that climatic suitability plays a consistent role in structuring nosemosis dynamics across Turkish apicultural systems. The present study extends these regional observations by integrating hierarchical and nonlinear modeling within a spatially explicit framework, thereby quantifying the magnitude of climatic associations rather than merely reporting prevalence patterns. In summary, our findings indicate that moisture availability is a strong climatic correlate of nosemosis incidence under the field conditions

sampled here, while recognizing that direct mechanistic attribution will require multi-stressor designs that jointly account for non-climatic determinants.

4.3. Nonlinear and thermal performance patterns

The significant quadratic temperature term in the mixed-effects model (LMM) provides evidence for a nonlinear temperature–infection relationship that persisted after controlling for province-level heterogeneity. Infection rates increased with temperature up to an intermediate range, after which marginal gains diminished, consistent with thermal performance theory. The GAM without random effects similarly detected strong nonlinear smooths (edf = 7.44 for temperature); however, these were substantially attenuated in the GAMM (edf ≈ 1), suggesting that much of the apparent GAM-detected nonlinearity is confounded by between-province variation. Importantly, the quadratic LMM term, which already included province as a random intercept, remained significant; this indicates that a degree of genuine nonlinear temperature response exists beyond spatial confounding, even if its magnitude is more modest than the pooled GAM smooth implies. Interpreting the GAMM result as evidence of linear climatic effects would be premature given the convergence of the quadratic LMM and the ecological plausibility of thermal performance curves for microsporidian pathogens (Fenoy et al., 2009). Future studies with larger spatial replication are needed to fully disentangle these effects. Thermal performance theory predicts unimodal pathogen responses to temperature, reflecting physiological constraints on replication and host–pathogen interactions (Mordecai et al., 2013). Laboratory evidence that *N. ceranae* replicates optimally within certain temperature ranges (Fenoy et al., 2009) is consistent with our field observations. This finding reinforces the importance of integrating nonlinear and hierarchical modeling approaches when evaluating ecological disease dynamics (Zuur et al., 2009). Taken together, the contrast between the GAM (edf ≈ 7–9) and the GAMM (edf ≈ 1) provides an important methodological caveat: the smooth functions estimated by the pooled GAM should not be interpreted as direct evidence of strong intrinsic nonlinear climatic responses, because once spatial heterogeneity is partitioned through province-level random effects, the residual within-province climatic response becomes nearly linear. We therefore distinguish between (i) a modest but ecologically interpretable nonlinear temperature response, which is supported by the quadratic LMM and is consistent with thermal performance theory, and (ii) the more pronounced nonlinearity observed in the pooled GAM smooths, which is at least partly attributable to between-province compositional differences rather than to a true within-colony functional response. Accordingly, the “thermal performance curve” descriptor is retained only in this more restricted, quadratic-LMM-based sense, and the strength of nonlinearity should not be inferred from the pooled GAM alone. Four mutually compatible mechanisms can plausibly account for why the addition of hierarchical random effects collapsed the GAM smooths to near-linearity (edf ≈ 1). First, climatic gradients are themselves spatially structured at the provincial scale (cooler, drier inland provinces such as Sinop versus warmer, more humid coastal provinces such as Rize and Trabzon), so much of the apparent climatic non-linearity in the pooled GAM is in fact a between-province compositional effect rather than a within-colony functional response. Second, this is a textbook case of spatial confounding: when the spatial random effect and the fixed-effect climatic gradient share the same spatial axis, partitioning the spatial structure necessarily absorbs the systematically varying component of the climatic effect. Third, the pooled GAM may be partially overfitting the across-province climatic gradient under the default thin-plate-spline penalty; consistent with this, restricted-maximum-likelihood (REML) smoothness selection in the GAMM penalizes the unstructured wiggliness more strongly. Fourth, the quadratic LMM is itself a constrained, low-edf representation of non-linearity, so the conclusion that the temperature response is

“non-linear at all” is robust only to the extent that the quadratic term in the LMM remains statistically significant, which it does ($t = 4.21$); however, this alone does not confirm a biologically interpretable thermal optimum, only that a symmetric low-order curvature is detectable in the data once spatial structure is accounted for. We therefore present the thermal-performance interpretation as a hypothesis that is supported by, but not confirmed by, our field data, and that requires independent validation in experimental thermal performance assays.

4.4. Hierarchical spatial structure

Approximately 18% of the total variance in infection prevalence was attributable to provincial differences. This moderate intraclass correlation indicates that the macroecological context contributes meaningfully to infection risk, while leaving the majority of variance (~82%) unexplained by province-level structure – a residual that is most likely shaped by other, unmeasured determinants such as colony strength, nutrition, beekeeping practices, pesticide exposure, and co-infections. Mixed-effects modeling is critical in nested ecological datasets, as it allows for proper variance partitioning and prevents inflated inference (Bates et al., 2015; Nakagawa and Schielzeth, 2013). The nonsignificant elevation effect further clarifies that altitude itself does not directly determine infection risk when climatic variables are explicitly modeled; rather, elevation likely operates indirectly through its influence on local temperature and humidity. Nationwide molecular surveys in Türkiye have revealed heterogeneous prevalence patterns across provinces (Ütük et al. 2016), supporting the interpretation that local ecological conditions shape the infection dynamics.

4.5. Spatial dispersion rather than clustering

Exploratory spatial analysis using Global Moran's I indicated a negative spatial autocorrelation ($I = -0.476$, $p = 0.043$), suggesting spatial dispersion rather than clustering at the provincial scale. Given that this analysis was based on only seven spatial units, this finding must be interpreted with considerable caution and is best viewed as hypothesis generating rather than confirmatory. In practical terms, if the dispersion pattern holds with finer-scale data, high-risk provinces are not expected to be surrounded by similarly high-risk neighbors. This suggests that infection patterns are structured by localized climatic suitability rather than contiguous epidemic spread. Similar spatial heterogeneity in nose-mosis prevalence has been observed in multi-year European monitoring studies (Gisder et al., 2010). Spatial dispersion may arise from microclimatic variations, landscape heterogeneity, differences in colony management, or stochastic colony-level processes. However, given the limited statistical power inherent to $n = 7$ spatial units, we refrain from drawing firm ecological inferences regarding the absence of contagion or the relative importance of local environmental conditions versus spatial spread at the provincial scale; such inferences should be revisited in future studies employing finer-scale (locality- or apiary-level) spatial designs and dedicated spatial regression models.

4.6. Methodological implications: beta mixed modeling

Because infection prevalence represents proportional data, the use of beta mixed-effects regression was methodologically appropriate. Gaussian assumptions can produce biased estimates when modeling bounded proportions (Cribari-Neto and Zeileis, 2010). The beta mixed model confirmed strong climatic effects while maintaining a hierarchical structure and improving distributional realism (Brooks et al., 2017). Five-fold cross-validation of the beta mixed-effects model yielded a mean RMSE of 0.109 on the proportional scale (approximately 10.9% on the percentage scale). RMSE values were consistent across folds, indicating stable predictive performance. Although this level of error reflects moderate predictive accuracy, it is ecologically realistic given the multifactorial nature of honeybee health systems, which are influenced

by nutrition, pesticide exposure, colony strength, and pathogen interactions (Doublet et al., 2015; Goulson et al., 2015). Thus, climatic variables explain a substantial proportion of infection variability; however, additional stressors likely contribute to unexplained variance.

4.7. Implications under climate change

Climate change is expected to alter regional temperature and humidity regimes, potentially reshaping host–pathogen interactions. Pollinator decline has been linked to interacting stressors, including climate variability (Goulson et al., 2015; Potts et al., 2010). Given the non-linear and threshold-like responses suggested by our data, it is plausible – though not directly tested in this study – that small climatic shifts could translate into disproportionate epidemiological effects under future climate scenarios. We emphasize that the present analysis did not incorporate downscaled climate projections, did not evaluate emission-pathway scenarios, and did not quantify temporal trends in the climatic predictors themselves; consequently, the following implications are offered as hypotheses for future, projection-based investigations rather than as inferential conclusions of the current observational dataset. Under these caveats, regions approaching the inferred thermal optimum, when combined with persistently high relative humidity, may hypothetically experience amplified infection prevalence as climates warm, while extreme heat events could in principle suppress spore viability beyond physiological limits. These findings underscore the need for climate-integrated surveillance and adaptive apicultural management strategies in Türkiye and other comparable climatic zones.

4.8. Limitations and future directions

Although this study integrates multiple advanced statistical approaches, several limitations warrant consideration. First, the spatial resolution remains at the provincial scale, as province-level modeling captures broad climatic gradients but may obscure finer-scale heterogeneity at the locality or apiary level. With only seven spatial units ($n = 7$), the power to detect complex spatial structures is inherently limited, and spatial inference, including the exploratory Moran's I result, should be interpreted cautiously. Future research incorporating locality-level spatial coordinates and spatial regression models (e.g., SAR or spatial error models) would provide more refined inferences regarding spatial contagion versus environmental filtering processes. Second, the present study focused exclusively on climatic drivers; integrating colony strength metrics, nutritional data, and pesticide exposure would enable the evaluation of multi-stressor interactions, as recommended in broader apian health research (Doublet et al., 2015). Finally, extending the four-year dataset to longer time series and incorporating climate projections would further strengthen interannual inference and predictive capacity. Future research should integrate climate projections with mechanistic host–pathogen models to forecast region-specific infection risks. Third, and most importantly for the interpretation of our findings, the fixed-effect structure was deliberately restricted to two climatic predictors (temperature and relative humidity); a wide range of established determinants of honeybee health and nose-mosis dynamics – including colony strength, brood-to-adult ratio, queen status, nutritional state (pollen and nectar availability), beekeeping practices (treatment regimes, hive movement, supplementary feeding), pesticide exposure (notably neonicotinoids), and co-infections with viruses (e.g., DWV, BQCV) or Varroa destructor – were not measured and could not be incorporated into the models. Consequently, the climatic variables modeled here should be regarded as explaining only a portion of the total variability in infection prevalence, and the statistical associations reported must not be interpreted causally. The substantial residual variance observed in the beta mixed-effects model ($\text{RMSE} \approx 0.109$) and the moderate cross-validation accuracy are consistent with this multifactorial reality. Future work should adopt a multi-stressor design in which climatic predictors are jointly modeled with colony-, apiary-, and

landscape-level covariates to enable proper attribution of the relative contributions of climatic versus non-climatic determinants of nosemosis risk.

5. Conclusion

In summary, nosemosis dynamics in the Eastern Black Sea region appear to be strongly associated with climatic variability. Humidity and temperature emerged as primary climatic correlates of infection prevalence, while the hierarchical spatial structure explains substantial variance. Both linear (LMM quadratic term) and nonlinear (GAM) analyses indicate a thermal performance pattern in the temperature response, although the attenuation of GAM smooths in the GAMM underscores the importance of simultaneously accounting for spatial heterogeneity and nonlinear responses. The exploratory Moran's I result is best regarded as hypothesis-generating only and requires confirmation with finer-scale (locality- or apiary-level) data before any spatial-structural conclusions can be drawn. This study provides a robust epidemiological framework for understanding climate–pathogen associations in Turkish apiculture systems by integrating nonlinear modeling, hierarchical structure, beta regression, and spatial statistics. Because the analysis was restricted to climatic predictors, the relationships reported here describe statistical associations rather than causal links, and the unexplained variance is likely shaped by unmeasured factors such as colony strength, nutrition, beekeeping practices, pesticide exposure, and co-infections; multi-stressor designs incorporating these covariates at finer spatial resolution will be needed to test, refine, and extend the framework developed here.

CRedit authorship contribution statement

Çağrı Bekircan: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Onur Tosun:** Resources, Investigation, Data curation. **Muammer Kurnaz:** Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation.

Author contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Onur TOSUN, Çağrı BEKİRCAN and Muammer KURNAZ. The first draft of the manuscript was written by Çağrı BEKİRCAN and all authors commented on previous versions of the manuscript. All authors read and approved of the final manuscript.

Consent for publication

This study does not involve any individual person's data in any form (including images, videos, or personal details). Therefore, consent for publication is not applicable. All data presented are derived from experimental procedures and do not contain identifiable personal information.

Ethical approval

This study did not involve human participants, personal data, or animal experimentation requiring formal ethical clearance. Accordingly, no official ethical approval was necessary. The research was conducted in full compliance with institutional guidelines and international standards, and no procedures or evaluations of an unethical nature were performed at any stage.

Funding

The authors declare that no funds, grants, or other support were

received during the preparation of this manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Any opinions, findings, and conclusions expressed in this manuscript are solely those of the authors and do not necessarily reflect the views of the affiliated organizations. We gratefully acknowledge the Turkish State Meteorological Service (Meteoroloji Genel Müdürlüğü, TSMS) for providing the monthly mean temperature and relative humidity records used in this study, and the local beekeepers of the seven Eastern Black Sea provinces (Artvin, Rize, Trabzon, Giresun, Ordu, Samsun, and Sinop) for granting access to their colonies and for their cooperation throughout the four-year sampling campaign.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.vetpar.2026.110835](https://doi.org/10.1016/j.vetpar.2026.110835).

Data Availability

All raw data generated and analyzed during this study have been included in the supplementary material accompanying this manuscript. The supplementary files provide full access to the datasets to ensure transparency, reproducibility, and further scholarly use.

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